# Technical Note: Network-Topology-Aware Accessibility Modelling for the MO-IHLNDP Synthetic Dataset

Phu-Quy Le-Tang[1], Quang-Vu Nguyen[1], and Sunarin Chanta[2]

[1] The University of Danang – Vietnam-Korea University of Information and Communication Technology
[2] King Mongkut's University of Technology North Bangkok
quyltp.22git@vku.udn.vn, nqvu@vku.udn.vn, sunarin.c@itm.kmutnb.ac.th

**Abstract.** This technical note documents a methodological correction to the synthetic dataset used in the MO-IHLNDP case study for Central Vietnam. The original dataset generated arc-accessibility indicators under a complete-graph assumption, which conflates link existence with link disruption. We replace this with a two-stage model: a Delaunay triangulation defines the planar network topology, and BFS path-reachability extends accessibility to non-adjacent node pairs. Transport costs for non-adjacent pairs are updated to shortest-path distances through the planar network. This note describes the original methodology, its limitations, the corrective model, and the implementation plan.

**Keywords:** Synthetic dataset · Accessibility modelling · Delaunay triangulation · Path reachability · Humanitarian logistics

## 1 Network-Topology Limitation and Planned Correction

### 1.1 Current Methodology

The synthetic dataset generates the arc-accessibility indicator $a_{uvms}$ under a **complete-graph assumption**: for every ordered pair $(u, v) \in V \times V$, across all modes $m$ and scenarios $s$, a disruption probability is computed as

$$p_{uvs}^{\text{road}} = \min\bigl(0.97,\ \beta_s \cdot \bar{r}_{uvs}\bigr), \qquad \bar{r}_{uvs} = \tfrac{r_{us}+r_{vs}}{2}, \tag{1}$$

where $\beta_s \in \{0.25, 0.55, 0.88\}$ scales with scenario severity (mild, severe, extreme). Water mode is enabled for a pair $(u, v)$ when both endpoints exceed a flood-risk threshold ($r_{us} > 0.30$ and $r_{vs} > 0.30$); air remains universally accessible. Transport costs and travel times are estimated via Haversine distances adjusted for mode-specific tortuosity and terrain correction factors, computed independently for all $N^2$ pairs.

This formulation is aligned with prior two-stage stochastic relief logistics models [**?**, **?**] that adopt complete-graph networks for analytical tractability.

### 1.2 Identified Limitation

The complete-graph assumption conflates **link existence** with **link disruption**: it assigns a disruption probability to arcs that have no corresponding physical road or waterway in Central Vietnam's geography. Two consequences arise.

*First*, nodes on the convex boundary of the study region receive artificially high connectivity. Their pairwise risk average $\bar{r}_{uvs}$ is diluted by distant, lower-risk neighbours, so "belt" arcs along the convex hull remain accessible even under extreme flooding. This contradicts real infrastructure patterns along the narrow coastal corridor of the Quang Nam – Da Nang – Thua Thien-Hue provinces, where coastal roads are typically the first to be inundated or cut [**?**].

*Second*, transport costs for non-adjacent node pairs underestimate true distances: direct Haversine costs ignore mandatory detours through the inland mountain passes of the Truong Son range, whose terrain factor $\tau(\ell) \in [1.0, 1.8]$ substantially inflates effective travel distance for nodes separated by highland terrain.

### 1.3 Proposed Correction: Two-Stage Topology-Aware Model

We replace the complete-graph with a **two-stage network-topology-aware accessibility model**, implemented in `data_generate_cv.py`.

1. **Planar base graph.** A Delaunay triangulation of all node coordinates defines the set of *directly adjacent* arcs $\mathcal{E} \subset V \times V$, yielding $|\mathcal{E}| \approx 3N$ edges (346 edges for $N = 132$) — a sparse, geographically plausible planar network. The triangulation is computed once per instance via `scipy.spatial.Delaunay`.
2. **Edge-level disruption.** The risk-based disruption (Eq. 1) and water-threshold logic are applied exclusively to $\mathcal{E}$, producing a scenario-specific accessible subgraph $G_s^m = (V, \mathcal{E}_s^m)$ for each mode $m$. Non-adjacent arcs are initialised to zero (blocked) and are never assigned a disruption draw.
3. **Path-reachability extension.** For all non-adjacent pairs $(u, v)$, we set $a_{uvms} = 1$ if and only if $v$ is reachable from $u$ in $G_s^m$ via BFS; otherwise $a_{uvms}$ remains zero. This preserves the binary structure of the existing constraint set (Constraints 10–11 in the original model) while capturing realistic route detours through intermediate nodes.
4. **Shortest-path transport costs.** For road (mode 0) and water (mode 1), the direct Haversine cost is retained for Delaunay-adjacent pairs; for non-adjacent pairs the cost matrix entry $C_{uvm}$ is overwritten with the Dijkstra shortest-path cost through $\mathcal{E}$, so that model decisions correctly reflect the cost of the least-expensive multi-hop route. Air (mode 2) retains direct Haversine costs, as helicopters fly straight lines.

This correction ensures that under extreme scenarios, isolated coastal nodes genuinely lose road access rather than retaining it through spurious direct arcs, and that water transport's compensatory role along the Thu Bon and Huong rivers is geographically grounded.

### 1.4 Quantitative Impact

Table 1 reports the number of non-Delaunay node pairs that carry a road-accessibility value of one (i.e., path-reachable) under the corrected model for CV-Large ($N = 132$, 8 300 non-adjacent pairs).

**Table 1.** Path-reachable non-adjacent node pairs (road mode, CV-Large).

| Scenario | Reachable pairs | Fraction |
|---|---|---|
| Mild | 6 794 | 81.9% |
| Severe | 6 794 | 81.9% |
| Extreme | 6 016 | 72.5% |

Under the original complete-graph model, all $N(N-1)/2 = 8\,646$ non-diagonal pairs received independent disruption draws; under the corrected model, only the 346 Delaunay edges receive draws, and path-reachability determines the rest. The extreme scenario shows a meaningful reduction of 778 additional unreachable pairs compared with mild/severe, correctly capturing the greater network fragmentation caused by large-scale flooding.

### 1.5 Implementation Plan

Algorithm 1 summarises the corrected accessibility generation procedure, replacing lines 483–577 of `src/scripts/data_generate_cv.py`.

Transport costs for non-adjacent pairs are updated by running Dijkstra from every source node through $\mathcal{E}$ (road and water modes only), replacing the direct Haversine entry with the shortest-path distance. The visualiser (`visualizer/dataset_view.py`) requires no changes: the Delaunay edge display will now correctly reflect the corrected data topology once the instance JSON files are regenerated.

## References


1. F. Barzinpour and V. Esmaeili, "A multi-objective relief chain location distribution model for urban disaster management," *International Journal of Advanced Manufacturing Technology*, vol. 70, no. 5–8, pp. 1291–1302, 2014.
2. S. Tofighi, S. A. Torabi, and S. A. Mansouri, "Humanitarian logistics network design under mixed uncertainty," *European journal of operational research*, vol. 250, no. 1, pp. 239–250, 2016.
3. M. T. Nguyen, Z. Sebesvari, M. Souvignet, F. Bachofer, A. Braun, M. Garschagen, U. Schinkel, L. E. Yang, L. H. Nguyen, V. Hochschild, A. Assmann, and M. Hagenlocher, "Understanding and assessing flood risk in Vietnam: Current status, persisting gaps, and future directions," *Journal of Flood Risk Management*, vol. 14, no. 2, p. e12689, 2021.

**Algorithm 1** Topology-aware accessibility generation (per scenario $s$)

**Require:** Node coordinates coords, scenario risk vector $\mathbf{r}_s$, disruption parameter $\beta_s$
**Ensure:** Accessibility tensor $a[m][u][v]$ for $m \in \{0, 1, 2\}$

$\mathcal{E} \leftarrow$ DelaunayEdges(coords) ▷ $\approx 3N$ undirected edges
$a[m][u][v] \leftarrow 0$ for all $m, u, v$ ▷ default: all blocked
**for all** $(u, v) \in \mathcal{E}$ **do**
  $\bar{r} \leftarrow (r_{us} + r_{vs})/2$
  $a[0][u][v] \leftarrow a[0][v][u] \leftarrow \mathbf{1}[\text{Uniform}(0, 1) \geq \min(0.97,\ \beta_s \cdot \bar{r})]$ ▷ road
  $a[1][u][v] \leftarrow a[1][v][u] \leftarrow \mathbf{1}[r_{us} > 0.30 \wedge r_{vs} > 0.30]$ ▷ water
  $a[2][u][v] \leftarrow a[2][v][u] \leftarrow 1$ ▷ air
**end for**
**for all** $m \in \{0, 1, 2\}$ **do**
  Build adjacency list $\text{adj}_m$ from $\{(u, v) \in \mathcal{E} : a[m][u][v] = 1\}$
  **for all** src $\in V$ **do**
    visited $\leftarrow \text{BFS}(\text{adj}_m, \text{src})$
    **for all** dst $\in$ visited, dst $\neq$ src **do**
      $a[m][\text{src}][\text{dst}] \leftarrow 1$
    **end for**
  **end for**
**end for**